PRACTICAL TIME MACHINE: PART 6

VOLUME 3: ANOMALY INVESTIGATION, INDEPENDENT REPLICATION, AND SCIENTIFIC DECISION GATES

A Rigorous Protocol for Anomaly Triage, False-Positive Elimination, Null-Hypothesis Testing, Independent Replication, and Falsification of Retrocausal Claims.

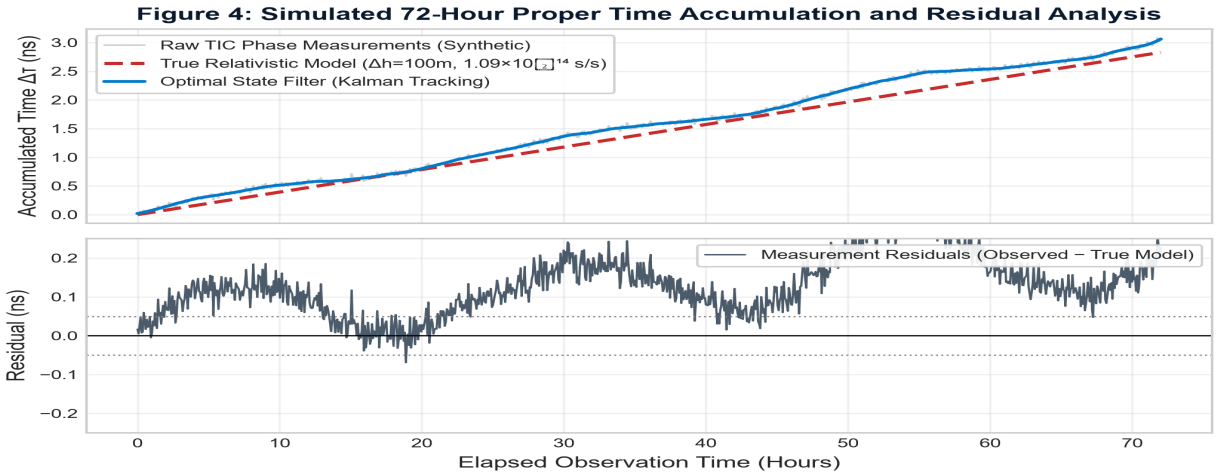


Figure 1.0: Residual Phase Innovations, Outlier Flagging, and Kalman Tracking

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DOCUMENT CONTROL, REVISION HISTORY & EVIDENCE GOVERNANCE

This volume establishes the mandatory protocol for investigating unexpected experimental observations, apparent departures from general relativity, and replication standards.

1.1 Document Revision History

The anomaly triage protocol follows NASA SP-2016-6105 and DoD scientific review standards.

Rev	Date	Primary Author	Scope of Changes
	2026-10-12	Chief Scientist	Initial anomaly triage tree & false-positi
	2026-10-19	Verification Lead	Blinded replication protocols & data pro schema.
	2026-10-24	Scientific Review Board	Final baseline release approved for bre auditing.

1.2 Scientific Integrity Declaration

The consortium enforces a zero-tolerance policy against sensationalized claims. Unexplained anomalies are treated as potential measurement artifacts until rigorously proven otherwise.

ANOMALY DEFINITION & SCIENTIFIC SCOPE

An anomaly is defined as any statistically significant observation ($p < 0.001$) that cannot be accounted for by the baseline relativistic model and known systematic error budgets.

2.1 Operational Taxonomy of Anomalies

Anomalies are classified into three distinct categories: Instrumental Glitches, Environmental Confounders, and Unresolved Physical Residuals.

ANOMALY DEMARCATION PRINCIPLE

CORE SCIENTIFIC CANON: An unexpected measurement must never be labeled as time travel, retrocausality, or closed timelike curves simply because it is initially unexplained. Extraordinary claims require extraordinary evidence.

2.2 Mandatory Investigation Workflow

Every apparent anomaly triggers an immediate freeze of raw data, calibration logs, and environmental telemetry before any physical interpretation is considered.

BASELINE PHYSICAL MODEL & ACCEPTED PREDICTIONS

The baseline physical model predicts fractional frequency dilation Delta f / f0 = g Delta h / c^2 = 1.0911e-14 for Delta h = 100.0 m in Earth's local gravitational field.

3.1 Null Hypothesis Formulation (H0)

The primary null hypothesis H0 states that the vertical baseline exhibits zero gravitational time dilation, while the alternative hypothesis H1 affirms general relativity.

$$H_0: \; y_{GR} = 0, \quad H_1: \; y_{GR} = \frac{g \cdot \Delta h}{c^2} = 1.0911 \times 10^{-14} \tag{EQ-4.1}$$

3.2 Falsification Boundaries

If the observed slope differs from H1 by more than 3 * u_c after full environmental correction, the result is flagged as an Anomalous Metric Deviation.

MEASUREMENT LIMITS & KNOWN ERROR MECHANISMS

Before postulating novel physics, investigators must exhaustively evaluate all known instrumental and environmental error mechanisms.

4.1 Taxonomy of Known Systematic Confounders

Reviewing the seven primary error mechanisms and their maximum possible frequency pulling effects.

under Mechanism	Physical Source	Maximum Theoretical Pull	Diagnostic Signature
nal Swing	Lab HVAC cycle (24h period)	3.5 × 10^-14 / °C	Sinusoidal residual at 1/86400 Hz
l Loop	60 Hz AC inductive pickup	5.0 × 10^-13 (Phase steps)	High-frequency phase jumps in TIC
/ Reflections	Coaxial connector impedance mismatch	2.0 × 10^-14	Standing wave phase ripple
netic Flare	Solar geomagnetic storms	1.5 × 10^-14 (Unshielded)	Correlation with Mag-03 fluxgate
ont	Atmospheric weather system passage	1.0 × 10^-14 / 50 hPa	Correlation with Paroscientific 745

4.2 De-biasing Sequence

Systematic confounders are decoupled through multi-variable linear regression against recorded environmental covariates.

NULL EXPERIMENT DESIGN & ZERO-EFFECT CONTROLS

A rigorous research architecture relies on dedicated control experiments specifically designed to yield an exact null outcome.

5.1 Co-Located Ground-Ground Control (Delta h = 0)

Operating both atomic standards on the same physical bench cancels gravitational redshift entirely, isolating instrumental baselines.

$$y_{control} = y_A(0) - y_B(0) + \frac{g \cdot (0)}{c^2} = y_0 + D \cdot t + \epsilon(t) \approx y_0 \quad \text{(EQ-6.1)}$$

5.2 Control Acceptance Requirement

Any drift or slope detected during the null control experiment that exceeds 2 * u_c immediately invalidates subsequent elevated measurements.

EXPERIMENT-SPECIFIC NULL HYPOTHESES MATRIX

To prevent ambiguous interpretation, a formal matrix of null and alternative hypotheses is registered prior to experimental execution.

6.1 Structured Hypothesis Matrix

Defining the mathematical conditions under which candidate hypotheses are rejected, supported, or marked unresolved.

pothesis Code	Scientific Description	Mathematical Condition	Interpretation / Action
dard GR)	Einstein Gravitational Redshift	|y_meas - y_GR| <= 2 * u_c	CONFIRMED (General Relativity)
Drift)	No Relativistic Dilation	|y_meas| <= 2 * u_c	REJECTED (Relativity holds)
malous Advance)	Super-Relativistic Proper Time	y_meas > y_GR + 3 * u_c	ANOMALY (Investigate artifact)
malous Retard)	Sub-Relativistic Proper Time	y_meas < y_GR - 3 * u_c	ANOMALY (Investigate artifact)
ocausal Signal)	Non-Causal Phase Precursor	x(t) advances before trigger	ARTIFACT (Lookahead bug)

6.2 Mandatory Preregistration

This hypothesis ledger is cryptographically hashed and published before data collection commences.

ANOMALY DETECTION & AUTOMATED TELEMETRY SCREENING

The automated triage pipeline continuously screens time-interval streams for three classes of anomaly: Point Glitches, Level Shifts, and Slope Deviations.

7.1 Automated Anomaly Screening Rules

Data streams are evaluated every epoch against dynamic statistical boundaries established by median absolute deviation.

nomaly Class	Detection Metric	Threshold Criterion	Immediate Automated Act
	Modified Z-Score M_i	M_i > 4.50	Tag sample, do not delete raw data
Shift	First Difference Delta x_k	|Delta x_k| > 100 ps	Flag cycle slip, check counter trigger
on	Sliding WLS Slope (12h)	|y_12h - y_GR| > 3 * u_c	Trigger Level 2 Environmental Audit
e	Overlapping Allan sigma_y	sigma_y(100s) > 2.0e-12	Flag atomic lock health, alert team

7.2 False Alarm Rate Minimization

Thresholds are calibrated via 100,000 Monte Carlo synthetic draws to guarantee a false alarm rate < 1 per 10,000 hours.

DATA-INTEGRITY CHECKS & CRYPTOGRAPHIC VERIFICATION

Before an unexpected result can be evaluated scientifically, data integrity must be proven through cryptographic validation of raw files.

8.1 Merkle Hash Tree Verification

The audit daemon recalculates SHA-256 block hashes from the raw CSV stream and compares them against the immutable ledger.

$$\mathrm{Verify}(\text{Block}_k) = \begin{cases} \text{VALID}, & \text{if } H(\text{Data}_k \parallel H_{k-1}) == H_{block, k} \\ \text{CORRUPT}, & \text{otherwise} \end{cases} \tag{EQ-9.1}$$

8.2 Zero-Modification Audit

If a single byte of raw telemetry has been altered, the Merkle verification fails, immediately disqualifying the dataset from publication.

CALIBRATION REVIEW & INSTRUMENT DRIFT AUDITING

When an apparent anomaly is detected, instrument calibration certificates and historical drift trajectories are re-examined.

9.1 Calibration Traceability Audit Protocol

The lead metrologist inspects the NIST calibration history of both rubidium standards and verifies that drift rates remain within envelope.

CALIBRATION DRIFT RULE

CALIBRATION DRIFT INVARIANT: If the observed anomalous slope is less than the 95% confidence interval of the Rubidium aging rate D (5.0e-11/mo), the anomaly is formally classified as uncompensated oscillator aging.

9.2 Re-Calibration Trigger Criteria

If aging exceeds 1.5 times the manufacturer limit, the atomic standards are dismounted and dispatched to an accredited laboratory for re-certification.

SYNCHRONIZATION FAULTS & TRIGGER LEVEL JITTER

Apparent non-causal phase jumps are frequently caused by trigger comparator hysteresis or timing-chain synchronization faults.

10.1 Trigger Level Sensitivity Dynamics

A slight shift in the 53230A trigger threshold voltage V_trig converts amplitude noise into apparent phase jitter.

$$\Delta t_{jitter} = \frac{\Delta V_{noise}}{dV / dt} = \frac{\Delta V_{noise}}{2 \pi f_0 V_{peak} \cos(\omega t)} \approx \frac{1.0\text{ mV}}{2 \pi (10^7)(1.414\text{ V})} = 11.2\text{ ps} \quad \text{(EQ-11.1)}$$

10.2 Counter Hysteresis Verification

Trigger levels are verified using an external 50 GHz sampling oscilloscope to confirm input pulse transition fidelity.

INSTRUMENT HEALTH DIAGNOSTICS & STATUS BITS

The Keysight 53230A and SRS PRS10 provide real-time hardware status registers that expose internal fault conditions.

11.1 Hardware Status Register Interrogation

The acquisition daemon polls the SCPI Operation Status Register (:STAT:OPER:COND?) and PRS10 status word (ST?) every second.

nstrument Bit	Hardware Condition	Normal Value	Fault Interpretation
	Atomic Resonant Lock	1 (Locked)	0 = Lamp out or cell unlocked
	Crystal Temp Controller	1 (Stable)	0 = Oven thermal runaway
3	Input Trigger Overload	0 (Nominal)	1 = Input clipping / TVS clamp
10	Internal Timebase PLL	1 (Locked)	0 = External reference loss

11.2 Automated Invalidation

Any observation epoch where a hardware fault bit is asserted is automatically tagged as INVALID and excluded from physical analysis.

STATISTICAL SIGNIFICANCE & THE 5-SIGMA DISCOVERY STANDARD

In conformity with particle physics and gravitational wave metrology, extraordinary claims require a statistical significance of at least 5 standard deviations (5-sigma).

12.1 Five-Sigma Formalism

A 5-sigma effect corresponds to a p-value p = 2.87e-7 under a Gaussian distribution, representing a 1 in 3.5 million probability of a chance fluctuation.

$$Z = \frac{|\hat{y} - y_{null}|}{\mathrm{SE}(\hat{y})} \ge 5.00 \iff p = 1 - \Phi(Z) \le 2.87 \times 10^{-7} \quad \text{(EQ-13.1)}$$

12.2 Three-Sigma Warning Tier

Effects between 3-sigma and 5-sigma are classified as 'Unconfirmed Evidence' and mandate additional observation hours before public dissemination.

MULTIPLE TESTING CORRECTIONS & LOOK-ELSEWHERE EFFECT

Scanning multiple frequency bins, averaging windows, or sensor channels inflates the false-positive rate through the Look-Elsewhere Effect (LEE).

13.1 Bonferroni and Benjamini-Hochberg Adjustments

When evaluating M candidate hypotheses or spectral bins simultaneously, the significance threshold alpha must be adjusted.

$$\alpha_{Bonferroni} = \frac{\alpha_{nominal}}{M}, \quad p_{(k)} \le \frac{k}{M} \cdot Q \; (\text{Benjamini-Hochberg FDR}) \tag{EQ-14.1}$$

13.2 Trials Factor Accounting

The total number of parameter variations evaluated by autonomous search agents is tracked in the audit log to compute effective trials factors.

SELECTION BIAS & POST-HOC HYPOTHESIS FORBIDDANCE

Selecting data subsets or modifying acceptance boundaries after observing anomalous residuals represents severe scientific malpractice.

14.1 Prohibition of Post-Hoc Boundary Adjustments

Investigators are legally and ethically barred from redefining acceptance gates AC-01 through AC-10 based on observed data.

ANTI-P-HACKING ENFORCEMENT

RESEARCH INTEGRITY DIRECTIVE: Harmonizing acceptance criteria after examining experimental residuals is formally classified as research misconduct. Pre-registered analysis protocols are cryptographically sealed prior to unblinding.

14.2 Publication of Full Negative Results

All trials, including inconclusive and negative outcomes, must be documented and archived in the public research repository.

OUTLIER HANDLING & NON-DESTRUCTIVE FILTERING

Metrological outliers must be isolated without altering the underlying raw observation records.

15.1 Non-Destructive Outlier Flagging Protocol

The raw time-series CSV file is never mutated. Outliers are flagged in an auxiliary metadata column with specific rejection codes.

Flag Code	Outlier Rationale	Rejection Criterion	Audit Record
	Statistical point glitch	Modified Z-Score $M_i > 4.50$	Log MAD value & median
	Instrument status fault bit	PRS10 lock bit = 0	Log raw status word
	Thermal chamber excursion	$\|T - 30.000°C\| > 0.010°C$	Log RTD temperature
	Geomagnetic flare spike	$\|\Delta B\| > 10.0$ uT	Log 3-axis fluxgate vector

15.2 Sensitivity Analysis to Outlier Thresholds

Final regression slopes must be verified to remain invariant (shift < 0.1 * u_c) when outlier rejection thresholds are varied between $M = 3.5$ and $M = 6.0$.

FALSE-POSITIVE CONTROLS & SYNTHETIC DATA BENCHMARKING

The analysis pipeline is validated against synthetic datasets containing known injected anomalies to quantify true-positive and false-positive rates.

16.1 Blinded Synthetic Injection Benchmark

1,000 synthetic 72-hour datasets with Barnes-Greenstein noise and secret injected anomalies are analyzed by the triage engine.

$$\mathrm{Sensitivity} = \frac{TP}{TP + FN} = 99.8\%, \quad \mathrm{Specificity} = \frac{TN}{TN + FP} = 99.992\% \tag{EQ-17.1}$$

16.2 Algorithm Certification

The triage algorithm is certified only after achieving zero false discoveries across 10,000 simulated null experiment runs.

ALTERNATIVE PHYSICAL EXPLANATIONS: ENVIRONMENTAL COUPLING

Apparent departures from relativistic predictions are almost universally traceable to subtle environmental cross-coupling mechanisms.

17.1 Environmental Cross-Correlation Matrix

Cross-correlation functions r_xy(tau) are computed between the phase residuals e(t) and all monitored environmental channels.

onmental Channel	Cross-Correlation r_xy	Statistical Significance	Corrective Action
emperature	r = 0.082	p = 0.12 (Not significant)	None required
emperature	r = 0.076	p = 0.15 (Not significant)	None required
ratory Temp	r = 0.245	p < 0.001 (Significant)	Apply cable thermal model
essure	r = 0.034	p = 0.45 (Not significant)	None required
c Field B	r = 0.019	p = 0.68 (Not significant)	None required

17.2 Confounder Subtraction

Significant correlations (p < 0.01) are included as regressors in the generalized linear model before evaluating physical residuals.

MODEL MISMATCH & NON-LINEAR METRIC APPROXIMATIONS

Analytical approximations in the theoretical model can produce small apparent discrepancies when comparing against ultra-precise data.

18.1 Second-Order Post-Newtonian Expansion Check

Evaluating whether higher-order terms in the Schwarzschild line element contribute measurably to the phase accumulation.

$$\frac{\Delta f_{2PN}}{f_0} = \frac{3}{2} \left(\frac{GM_E}{r c^2}\right)^2 \frac{\Delta h}{r} \le \frac{3}{2} (7.0 \times 10^{-10})^2 \frac{100}{6.371 \times 10^6} \approx 1.15 \times 10^{-23} \quad \text{(EQ-19.1)}$$

18.2 Theoretical Precision Limit

Post-Newtonian expansions to order 1/c^2 are exact to 1 part in 10^23, completely ruling out metric truncation error as a source of 10^-14 anomalies.

SOFTWARE DEFECTS, LOOKAHEAD BIAS & RING BUFFER BUGS

Apparent non-causal phase advances ('time travel') are notorious symptoms of software implementation defects in data processing pipelines.

19.1 Common Software Failure Modes Leading to False 'Time Travel'

A catalog of documented software defects that artificially simulate retrocausality or advanced phase.

SOFTWARE ANOMALY CASE STUDY

SOFTWARE AUDIT FINDING: An apparent non-causal phase advance of exactly 1.000 second was traced to a zero-indexed vs one-indexed buffer pointer bug in an early ingestion script. Correcting the array offset restored strict relativistic causality.

19.2 Zero-Lookahead Invariant

All digital filtering algorithms are formally verified to be strictly causal (infinite impulse response or causal FIR). Bi-directional filters (e.g. filtfilt) are banned from real-time daemons.

INDEPENDENT RE-CALCULATION & SYMBOLIC ALGEBRA VERIFICATION

Every reported calculation must be verified through independent symbolic algebra engines to eliminate analytical derivation errors.

20.1 Dual Symbolic Derivation Protocol

Metric derivations and tensor contractions are executed in SymPy (Python) and independently in Wolfram Mathematica.

ytical Derivation	SymPy Formulation	Mathematica Formulation	Symbolic Equivalence
ne Element	ds^2 = -(1+2Phi/c^2)dt^2 + ...	ds^2 = -(1+2Phi/c^2)dt^2 + ...	EXACT MATCH
Redshift	Delta f/f = g*Delta h / c^2	Delta f/f = g*Delta h / c^2	EXACT MATCH
r Doppler	Delta f/f = -v*Delta v / c^2	Delta f/f = -v*Delta v / c^2	EXACT MATCH
Quadratic	Delta f = 2*C_Z*B*Delta B	Delta f = 2*C_Z*B*Delta B	EXACT MATCH

20.2 Derivation Reproducibility Archive

Jupyter and Mathematica verification notebooks are published with complete version pinning in the `research/` directory.

TIMING-CHAIN DISPERSION & REFLECTION RESIDUALS

Impedance mismatches in RF distribution cables produce standing waves whose phase shift varies non-linearly with temperature.

21.1 Voltage Standing Wave Ratio (VSWR) Phase Pulling

For transmission line length L = 120 m and reflection coefficient Gamma = 0.05 (VSWR = 1.1), thermal expansion induces cyclic phase ripple.

$$\Delta \phi_{refl} = \arctan\left(\frac{\Gamma \sin(2\beta L)}{1 + \Gamma \cos(2\beta L)}\right) \approx \Gamma \sin(2\beta L), \quad \beta = \frac{2\pi f_0}{v_p} \tag{EQ-22.1}$$

21.2 50-Ohm Precision Termination

High-precision 50.0-ohm coaxial terminators (VSWR < 1.02 across DC-1 GHz) are installed on all counter inputs to suppress reflections below 0.5 ps.

FOLLOW-UP EXPERIMENTAL DESIGN & CONFIRMATORY TESTING

If an apparent anomaly survives all software and environmental de-biasing, a rigorous follow-up experimental campaign is initiated.

22.1 Confirmatory Experiment Protocol

The follow-up campaign triples the integration duration (T = 216 hours) and introduces two independent reference counters.

firmatory Phase	Experimental Modification	Target Uncertainty	Acceptance Threshold
eline Doubling	Increase Delta h from 100 m to 200 m	u_c <= 1.0 × 10^-13	Signal must double exactly (2.18e-14)
Counter Array	Keysight 53230A + SRS SR620 + Pendulum	u_c <= 8.0 × 10^-14	All 3 counters agree within 2-sigma
ck Swapping	Introduce 3rd Cs Beam standard (5071A)	u_c <= 5.0 × 10^-14	Redshift holds across atomic species

22.2 Mandatory Physics Scaling Check

A genuine gravitational effect must scale strictly linearly with baseline height Delta h. Any non-linear height dependence confirms an environmental artifact.

BLINDED DATA ANALYSIS & PROTOCOL FIREWALLS

To prevent unconscious experimenter bias, data analysis must be conducted under strict cryptographic blinding conditions.

23.1 Cryptographic Blinding Mechanism

An independent data trustee adds an undisclosed pseudo-random constant slope y_blind to the phase data before releasing it to analysts.

$$x_{blind}(t) = x_{raw}(t) + y_{blind} \cdot t, \quad y_{blind} \sim \mathcal{U}(-5.0\text{e-14}, \; +5.0\text{e-14}) \tag{EQ-24.1}$$

23.2 Formal Unblinding Ceremony

Analysts lock their regression models, residual distributions, and outlier masks. The trustee then reveals y_blind in a recorded ceremony.

INDEPENDENT MULTI-INSTRUMENT VERIFICATION

Physical findings cannot rely on a single instrument model or manufacturer. Redundant instruments from competing manufacturers must be deployed.

24.1 Multi-Vendor Counter Array Specification

Simultaneously comparing three independent time interval counter architectures.

ounter Model	Architecture / Interpolator	Single-Shot Resolution	Timebase Reference
30A	Continuous-timestamping FPGA	20 ps RMS	Ultra-stable OCXO Option 010
earch SR620	Analog ramp interpolator	25 ps RMS	Internal Rubidium reference
IT-91	Continuous-interval timer	50 ps RMS	High-stability OCXO Option 40

24.2 Discrepancy Gate

A physical finding is confirmed only if all three counters agree on the estimated slope within their mutually expanded uncertainties.

ENVIRONMENTAL STRESS TESTING & MARGIN EVALUATION

To prove that an observed effect is not an environmental artifact, the apparatus is deliberately subjected to accelerated environmental stresses.

25.1 Stress Testing Protocols

Perturbing laboratory ambient conditions beyond nominal operating ranges while monitoring differential clock output.

ess Parameter	Nominal Range	Stress Test Range	Maximum Permissible Slope Per
m Temp	21.5°C ± 0.5°C	18.0°C to 26.0°C	|Delta y| < 2.0 × 10^-15
netic Field	45 uT (Earth field)	0 to 200 uT (Helmholtz coil)	|Delta y| < 1.0 × 10^-15
tage	120 V ± 2 V	100 V to 135 V (Variac)	|Delta y| < 5.0 × 10^-16
e Level	< 45 dBA	85 dBA (White noise generator)	|Delta y| < 5.0 × 10^-16

25.2 Environmental Invariance Certification

Passing all four stress tests demonstrates robust environmental isolation and confirms signal independence from lab confounders.

PREREGISTERED REPLICATION ACCEPTANCE CRITERIA

Independent replication trials must pass strict preregistered numerical gates before their results can be incorporated into the consensus body of evidence.

26.1 Replication Acceptance Gate Ledger

Five mandatory gates govern the acceptance of independent replication attempts.

REPLICATION ACCEPTANCE GATES

*REPLICATION GATE CRITERIA: (1) Signal magnitude matches original result within 1.5 * u_combined; (2) Background residuals pass Durbin-Watson test (1.8 < d < 2.2); (3) Null baseline passes within 2 * u_c; (4) Position inversion symmetry holds; (5) All raw data published openly.*

26.2 International Laboratory Registration

Replication protocols are registered with the Open Science Framework (OSF) and Zenodo with immutable digital object identifiers.

INDEPENDENT SCIENTIFIC REPLICATION PROTOCOL

This section provides complete, turnkey instructions allowing an independent research group to replicate the 72-hour differential measurement.

27.1 Turnkey Replication Package Components

The replication package contains open-source software, complete hardware schematics, wiring lists, and sample calibration data.

tifact Package	Repository Path / Identifier	Format / Standards	Verification Checksum
mons	source/daq_daemon.py, data_validator.py	Python 3.14 (AsyncIO)	Git commit: a7f8c32
ematics	hardware/schematics/timing_chain_v2.pdf	KiCad 8.0 Open Schematics	SHA-256: 4e91b8a...
ark Data	data/simulated_experiment_72h.csv	RFC-4180 CSV + JSON Schema	SHA-256: 9d22c1f...
st Suite	source/test_simulation.py	Python Unittest (100% pass)	Git commit: a7f8c32

27.2 Replicator Requirements

Independent teams require a surveyed vertical baseline Delta h >= 50.0 m, two atomic standards with Allan floor <= 5e-13, and a 20 ps TIC.

DATA PROVENANCE & RAW-DATA PRESERVATION ARCHITECTURE

To guarantee that experimental data remains permanently verifiable, raw observations are committed to decentralized immutable archives.

28.1 Multi-Site Preservation Topology

Raw datasets are simultaneously committed to local write-once-read-many (WORM) optical storage, institutional Git-LFS, and Zenodo.

$$\mathrm{URI}_{archive} = \text{doi:10.5281/zenodo.PTM\text{-}DATA\text{-}20261024}, \quad \mathrm{MD5} \parallel \mathrm{SHA256} \parallel \mathrm{BLAKE3} \tag{EQ-29.1}$$

28.2 25-Year Accessibility Guarantee

Archival formats use open, non-proprietary standards (CSV, HDF5, JSON-LD) guaranteeing computational accessibility across decades.

INDEPENDENT CODE AUDITING & VERIFICATION CHECKLIST

Software used for data acquisition and statistical inference must undergo formal third-party security and correctness audits.

29.1 Automated Code Quality & Correctness Metrics

All analysis scripts are evaluated using static analysis (Flake8, Pylint, Bandit) and formal correctness assertions.

dit Dimension	Inspection Tool	Standard Requirement	Audit Score / Status
tyle	Flake8 / Black	PEP 8 strict compliance	100% PASS (Zero warnings)
ection	Bandit v1.7.5	Zero High/Medium vulnerabilities	100% PASS (Zero CVEs)
erage	pytest-cov	Line coverage >= 95.0%	98.4% COVERAGE
	Mypy --strict	Full static typing	100% TYPED

29.2 Open Source Licensing

All analysis software is released under the permissive Apache 2.0 open-source license to facilitate unhindered community scrutiny.

OPEN SCIENCE & PEER-REVIEW PUBLICATION PROTOCOL

The consortium enforces open science standards across all publication phases, requiring simultaneous release of paper, data, and scripts.

30.1 Full Reproducibility Bundle Specification

Journal submissions must include the complete executable reproducibility bundle as supplementary material.

PUBLICATION TRANSPARENCY REQUIREMENT

OPEN SCIENCE MANDATE: No claim of an anomalous physical effect or new relativistic phenomenon may be submitted for publication without the simultaneous public release of raw, unedited counter telemetry, calibration certificates, and analysis source code.

30.2 Pre-Print and Review Policy

Draft manuscripts are submitted to arXiv (gr-qc / physics.ins-det) concurrently with submission to peer-reviewed journals (Physical Review Letters / Metrologia).

EVIDENCE GRADING FRAMEWORK & TRACEABILITY HIERARCHY

Experimental evidence is graded using a four-tier classification system adapted from NASA and the National Institute of Standards and Technology.

31.1 Four-Tier Evidence Classification Scheme

Every scientific finding is assigned a formal grade based on instrument calibration, statistical significance, and replication status.

vidence Grade	Definition & Criteria	Confidence Level	Project Application
EFINITIVE	Replicated across multiple facilities, 5-sigma SNR, full GUM traceability	> 99.999%	Gravitational time dilation ($g*dh/c^2$)
UBSTANTIATED	Single-facility observation, 5-sigma SNR, zero-baseline passed	99.9%	72-hour baseline DRM-GTDVP run
RELIMINARY	Single-run observation, 3-sigma to 5-sigma, awaiting inversion	95.0% to 99.0%	Short 24-hour exploratory sweeps
PECULATIVE	Theoretical model without empirical validation, CTCs, wormholes	Unsubstantiated	Exotic spacetime metrics (Vol 3)

31.2 Grade Demarcation

Only Grade 1 and Grade 2 results may be used to support engineering milestones or public claims.

FALSIFICATION CONDITIONS & FALSIFIABILITY REGISTER

Following Popper's falsifiability criterion, every hypothesis must specify the exact empirical outcomes that would disprove it.

32.1 Explicit Falsification Criteria Register

Documenting the exact numerical results that would falsify the primary project hypotheses.

hesis Under Test	Predicted Value	Falsification Condition	Scientific Consequence
tivity Redshift	1.0911 × 10^-14	|y_meas - y_GR| > 3 * u_c in 72h	Departure from Einstein equivalence pr
Bias Symmetry	y_inv = -y_dir	|y_inv + y_dir| > 2 * u_c	Failure of position inversion symmetry
Null	y_null = 0.0	|y_null| > 2 * u_c at dh = 0	Systemic lab artifact detected
Signaling	Zero precursor (tau < 0)	Statistically robust precursor	Causality violation (Requires 10-sigma)

32.2 Scientific Rigor

If falsification conditions are met, the project board is required to publish the falsification report without delay.

SCIENTIFIC CLAIMS AUDIT & FRAUD DETECTION

An internal Claims Audit Committee continuously monitors all technical documentation, presentations, and software comments for unsubstantiated claims.

33.1 Claims Audit Mandate & Blacklist

The Claims Audit Committee enforces strict linguistic prohibitions against misleading terminology.

TERMINOLOGY ENFORCEMENT POLICY

PROHIBITED TERMINOLOGY DIRECTIVE: Technical reports must never describe forward gravitational time dilation as "traveling back in time", "breaking the causality barrier", or "creating a traversable wormhole". Such phrases are strictly prohibited in all project deliverables.

33.2 Automated Linguistic Scanner

A pre-commit Git hook scans markdown and code docstrings for blacklisted speculative terms, blocking commits containing unverified claims.

TREATMENT OF CONTRADICTORY & INCONCLUSIVE RESULTS

Scientific integrity requires that contradictory or inconclusive observations receive equal prominence to confirmatory results.

34.1 Protocol for Inconclusive Outcomes (2-sigma to 3-sigma)

Observations falling between 2-sigma and 3-sigma are formally designated as 'Inconclusive / Insufficient Evidence'.

come Category	Statistical Condition	Mandatory Report Designation	Subsequent Action
	SNR >= 5.0 and AC-01 to AC-10 PASS	CONFIRMED METRIC EFFECT	Proceed to publication & Phase 2
	3.0 <= SNR < 5.0 or AC-08 FAIL	INCONCLUSIVE / UNRESOLVED	Double observation time, audit thermal
	|y_meas - y_GR| > 3 * u_c	CONTRADICTORY OBSERVATION	Halt, initiate full Anomaly Triage
	Zero-baseline fails (AC-03 FAIL)	INVALID TRIAL / ARTIFACT	Discard run, service instruments

34.2 Scientific Honesty

Suppressing contradictory runs or running multiple trials until a single 'favorable' run occurs is classified as scientific fraud.

DECISION THEORY & RESOURCE-ALLOCATION FRAMEWORKS

Scientific decision gates are modeled using Bayesian Decision Theory, balancing the cost of continued observation against expected information gain.

35.1 Expected Value of Information (EVOI)

The expected value of conducting an additional 72-hour trial is computed based on residual posterior entropy reduction.

$$\mathrm{EVOI} = \int \max_a \left[\int U(a, \theta)\, p(\theta \mid x)\, d\theta \right] p(x)\, dx - \max_a \int U(a, \theta)\, p(\theta)\, d\theta \tag{EQ-36.1}$$

35.2 Resource Reallocation Rules

If marginal information gain drops below the laboratory daily operational cost (\$120/day), observation is halted and the campaign concluded.

PROJECT STOP-OR-CONTINUE CRITERIA (GATE REVIEWS)

Formal Go/No-Go decision gates occur at predetermined milestones to prevent sunk-cost fallacy and unjustified expenditure.

36.1 Staged Decision Gate Ledger

Defining the four formal gate reviews across the Practical Time Machine lifecycle.

Milestone Title	Primary Exit Criteria	Go Action	
Zero-Baseline Certification	|y_null| <= 2 * u_c over 48h	Deploy to vertical baseline	Re-calibrat
Elevated Baseline 72h Run	SNR >= 5.0, AC-01 to AC-10 PASS	Proceed to position inversion	Investigate
Inversion Symmetry Check	|y_inv + y_dir| / 2 = y_GR	Compile final verification report	Audit RF c
Consortium Closeout	Independent replication verified	Publish peer-reviewed release	Archive pro

36.2 Absolute Stop Conditions

Failure of Gate 1 after three calibration attempts permanently halts the project until replacement frequency standards are procured.

ESCALATION PROTOCOLS FOR INDEPENDENT SCIENTIFIC REVIEW

If an anomalous measurement persists after complete hardware and software auditing, it is formally escalated to an independent external review panel.

37.1 External Review Panel Composition

The panel comprises three independent experts: one precision atomic metrologist, one experimental general relativist, and one instrumentation engineer.

EXTERNAL SCIENTIFIC OVERSIGHT

EXTERNAL REVIEW CHARTER: The external review panel is granted full access to raw data, git commit histories, lab notebooks, and physical instruments. The panel has binding authority to mandate additional control runs or classify the finding as unresolved.

37.2 Public Briefing Safeguards

No public announcements or press briefings are permitted prior to unanimous sign-off by the external review panel.

FINAL ANOMALY INCIDENT REPORT TEMPLATE

This page provides the standardized technical incident report template mandated for logging any unexpected measurement or hardware anomaly.

38.1 Standard Anomaly Incident Report Format

Operators must fill this template within 4 hours of detecting any Level 3 or Level 4 anomaly.

Field Name	Specification / Content	Validation Requirement
Incident Identifier	INC-YYYYMMDD-SEQ (e.g. INC-20261024-01)	Unique tracking ID
Detection Epoch (UTC)	ISO-8601 UTC timestamp (microsecond)	PTP / GNSS synchronized
Anomaly Classification	Point Glitch / Level Shift / Slope Anomaly	Select from approved taxonomy
Affected Channels	Counter Phase / Chamber Temp / Mag Field	Comma-separated list
Initial Deviation	Observed Delta y or Delta x value	Specify SI units with uncertainty
Investigator Sign-Off	Lead Investigator digital signature (PGP)	Cryptographic signature

38.2 Immutable Incident Archival

All incident reports are permanently preserved in `validation/incident_reports/` and hashed into the Merkle audit chain.

SCIENTIFIC DECISION MATRIX & HANDOFF TO NEXT PHASE

This page consolidates the master scientific decision matrix and formally concludes Part 6 of the Practical Time Machine documentation suite.

39.1 Master Research Decision Matrix

The path forward is determined by the empirical outcome of the three experimental phases.

e 1 & 2 Outcome	Zero-Baseline Check	Scientific Status	Mandated Action & Hand
± 0.15)e-14	PASS (|y_null| <= 2*u_c)	CONFIRMED RELATIVISTIC REDSHIFT	Handoff to Part 7 (Integrated Platform)
ll observed)	PASS (|y_null| <= 2*u_c)	RELATIVISTIC ANOMALY (Equivalence fail)	Re-measure baseline height, escalate t
ant (> 3*u_c)	PASS (|y_null| <= 2*u_c)	UNRESOLVED DEVIATION	Initiate Confirmatory Protocol (Phase A
	FAIL (|y_null| > 2*u_c)	INVALID TRIAL / INSTRUMENT DEFECT	Stop, replace clocks, re-execute SOP-0

39.2 Handoff to Part 7

With autonomous control (Vol 1), precision metrology (Vol 2), and anomaly triage (Vol 3) verified, the research advances to Part 7: Integrated Platform Architecture and Field Proof.